

Mathematics for Data Science

Exercise Sheet – Lecture 1

Vector Spaces, Subspaces, Linear Maps, Rank and Kernel

M1 – AI / Data Science

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Unless stated otherwise, all vector spaces are real vector spaces.

Part 1 – Course questions

Part 1(a) – Definitions

Question 1 (Euclidean space). In machine learning, a data point is often represented by a vector

$$x = (x_1, \dots, x_d) \in \mathbb{R}^d.$$

- (a) Explain why $\mathbb{R}^d$ is a vector space. Which operations are involved?
- (b) Explain why the following set S is a subspace of $\mathbb{R}^d$.

$$S = \left\{ x \in \mathbb{R}^d \left| \sum_{j=1}^d x_j = 0 \right. \right\}.$$

- (c) Give an example of a subset of $\mathbb{R}^d$ that is not a subspace, and explain which axiom fails.

Question 2 (Span). In dictionary learning, one represents a data vector $x \in \mathbb{R}^d$ approximately as

$$x \approx c_1 d_1 + \dots + c_k d_k,$$

where $d_1, \dots, d_k \in \mathbb{R}^d$ are called *dictionary atoms*.

- (a) Define the span of a family of vectors $d_1, \dots, d_k$.
- (b) Explain why $\text{span}(d_1, \dots, d_k)$ is always a vector subspace.
- (c) What does it mean, geometrically, for a data point x to belong to $\text{span}(d_1, \dots, d_k)$?
- (d) What can go wrong if the dictionary atoms are linearly dependent?

Question 3 (Basis, dimension, and embeddings). In ML, a categorical variable with K categories is sometimes represented by one-hot vectors

$$e_1 = (1, 0, \dots, 0), \quad \dots, \quad e_K = (0, \dots, 0, 1)$$

in $\mathbb{R}^K$.

- (a) Explain why $(e_1, \dots, e_K)$ is a basis of $\mathbb{R}^K$. What is the dimension of $\mathbb{R}^K$?

- (b) Categories may also be represented by dense vectors in $\mathbb{R}^r$, with $r \ll K$. These vectors are called *embeddings* (e.g., word embeddings). Explain, in linear algebra terms, what changes when one replaces one-hot vectors in $\mathbb{R}^K$ by embeddings in $\mathbb{R}^r$.

Question 4 (Linear maps, kernel, range, rank). Let $L : U \rightarrow V$ be a linear map.

- (a) Define $\ker(L)$, $\text{range}(L)$, and $\text{rank}(L)$.
 (b) State the rank-nullity theorem.
 (c) In a neural network, a dense layer without bias and without activation has the form

$$x \mapsto Wx.$$

Explain why this is a linear map.

- (d) Why is the map

$$x \mapsto Wx + b$$

generally not linear?

Part 1(b) – True/False questions

For each statement, decide whether it is true or false, and justify your answer.

T/F 1. Every subset of $\mathbb{R}^d$ containing the zero vector is a vector subspace.

T/F 2. If $S \subseteq \mathbb{R}^d$ is closed under addition and scalar multiplication, and S is nonempty, then S is a vector subspace.

T/F 3. The set

$$A = \left\{ x \in \mathbb{R}^d \mid \sum_{j=1}^d x_j = 1 \right\}$$

is a vector subspace of $\mathbb{R}^d$.

T/F 4. The span of a finite family of vectors is always a vector subspace.

T/F 5. If $v_1, \dots, v_k$ are linearly independent vectors in $\mathbb{R}^d$, then $k \leq d$.

T/F 6. If $v_1, \dots, v_d$ span $\mathbb{R}^d$, then they form a basis of $\mathbb{R}^d$.

T/F 7. If $L : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is linear and $k < d$, then L cannot be injective.

T/F 8. If $\ker(L) = \{0\}$, then L is injective.

T/F 9. If $L : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is linear and $\text{rank}(L) = k$, then L is surjective.

T/F 10. In a data matrix, if one feature is an exact linear combination of other features, then the corresponding columns are linearly dependent.

T/F 11. A PCA-type representation of a data point by coordinates in a low-dimensional subspace is an example of representing data using linear combinations of basis directions.

Part 2 – Computational exercise

Exercise 1. Let's assume that a ML system represents online products using three types of features:

$$x = (t, e, i) \in \mathbb{R}^4 \times \mathbb{R}^3 \times \mathbb{R}^3,$$

where:

- $t = (t_1, t_2, t_3, t_4) \in \mathbb{R}^4$ contains tabular numerical features;
- $e = (e_1, e_2, e_3) \in \mathbb{R}^3$ is a text embedding;
- $i = (i_1, i_2, i_3) \in \mathbb{R}^3$ is an image embedding.

The full representation space is therefore

$$V = \mathbb{R}^4 \times \mathbb{R}^3 \times \mathbb{R}^3.$$

Assume that, after preprocessing, admissible product (commercial item!) representations satisfy the following constraints:

$$t_2 = 100t_1, \quad t_4 = t_1 + t_3, \quad e_1 + e_2 + e_3 = 0,$$

and

$$i \in \text{span}(a, b), \quad a = (1, 0, 1), \quad b = (0, 1, 1).$$

NB: very simple image embeddings ;)

Let $S \subseteq V$ denote the set of all admissible representations.

1. What is the dimension of the ambient space V ?
2. Show that S is a vector subspace of V .
3. Describe separately:

$$S_t = \{t \in \mathbb{R}^4 \mid t_2 = 100t_1, t_4 = t_1 + t_3\},$$

$$S_e = \{e \in \mathbb{R}^3 \mid e_1 + e_2 + e_3 = 0\},$$

and

$$S_i = \text{span}(a, b).$$

Give an explicit basis of each of S_t , S_e , and S_i , and compute their dimension.

Now, we consider a model that does not use the full representation $x = (t, e, i)$. Instead, it keeps only the following four quantities (NB: feature extraction map Φ):

$$\Phi(x) = \begin{pmatrix} t_1 \\ t_3 \\ e_1 - e_2 \\ i_1 + i_2 + i_3 \end{pmatrix} \in \mathbb{R}^4.$$

4. Show that $\Phi : S \rightarrow \mathbb{R}^4$ is a linear map.
5. Determine whether Φ is injective.
6. Determine $\ker(\Phi)$ and compute $\dim(\ker(\Phi))$.
7. Use the rank-nullity theorem to compute $\text{rank}(\Phi)$.
8. Is Φ surjective onto $\mathbb{R}^4$?
9. Interpret $\ker(\Phi)$ in terms of information lost by the feature extraction map Φ .

The final prediction model is

$$g(x) = w^\top \Phi(x) + \beta,$$

where $w \in \mathbb{R}^4$ and $\beta \in \mathbb{R}$.

10. For fixed w , show that $x \mapsto w^\top \Phi(x)$ is linear on S .
11. Show that g is generally not linear when $\beta \neq 0$.
12. Explain why bias terms are useful in machine learning.

Part 3 – A bit of ML

Context

In computer vision, a grayscale image with d pixels can be represented as a vector

$$x = (x_1, \dots, x_d) \in \mathbb{R}^d,$$

where x_j is the intensity of pixel j .

A common objective is to replace the high-dimensional vector $x \in \mathbb{R}^d$ by a lower-dimensional vector

$$z \in \mathbb{R}^k, \quad k \ll d,$$

while keeping as much useful information as possible. The vector z is called a *low-dimensional representation* or an *embedding* of x .

In this exercise, we study a simplified version of this idea using only linear algebra from Lecture 1.

Preliminary definitions / before we define everything in the next lectures :)

For $x, y \in \mathbb{R}^d$, define the dot product by

$$x^\top y = \sum_{j=1}^d x_j y_j.$$

Assume that the vectors $u_1, \dots, u_k$ are orthonormal, meaning that

$$u_i^\top u_j = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

ML set-up

For any image vector $x \in \mathbb{R}^d$, define the map $E : \mathbb{R}^d \rightarrow \mathbb{R}^k$ as:

$$E(x) = \begin{pmatrix} u_1^\top x \\ u_2^\top x \\ \vdots \\ u_k^\top x \end{pmatrix} \in \mathbb{R}^k.$$

The map E is called an *encoder*. It maps the original image vector x to a lower-dimensional vector $z = E(x)$.

Define also the *decoder* map D by

$$D : \mathbb{R}^k \rightarrow \mathbb{R}^d, \quad D(z) = z_1 u_1 + \dots + z_k u_k.$$

It maps low-dimensional coordinates back to the original space.

Finally, we define the reconstruction of x by $\hat{x} = D(E(x))$.

Step 1 – Subspaces and dimension

1. Explain why the image vectors belong to the vector space $\mathbb{R}^d$.
2. Show that $S = \text{span}(u_1, \dots, u_k)$ is a vector subspace of $\mathbb{R}^d$.
3. What is $\dim(S)$? Justify your answer.
4. Explain geometrically why replacing $x \in \mathbb{R}^d$ by an element of S can be seen as dimensionality reduction.

Step 2 – Encoding as a linear map

5. Show that $E : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is linear.
6. What is the range of E ? Is E surjective?
7. Use the rank-nullity theorem to compute $\dim(\ker(E))$.
8. Describe $\ker(E)$ using the vectors $u_1, \dots, u_k$.
9. Give an interpretation for $\ker(E)$.

Step 3 – Decoding as a linear map

10. Show that $D : \mathbb{R}^k \rightarrow \mathbb{R}^d$ is linear.
11. What is $\text{range}(D)$?
12. Is D injective? Justify your answer.
13. Is D surjective onto $\mathbb{R}^d$ when $k < d$? Justify your answer.

Step 4 – Reconstruction

14. Write explicitly

$$\hat{x} = D(E(x)).$$

15. Show that

$$\hat{x} = \sum_{j=1}^k (u_j^\top x) u_j.$$

16. Show that $\hat{x} \in S$ for every $x \in \mathbb{R}^d$.
17. Suppose that $x \in S$. Show that $\hat{x} = x$.
18. Explain why $\hat{x}$ can be interpreted as the part of x that is kept by the low-dimensional representation.

Step 5 [Bonus] – Concrete example in dimension 5

Let $d = 5$ and $k = 2$. Define

$$u_1 = \frac{1}{\sqrt{2}}(1, 1, 0, 0, 0), \quad u_2 = \frac{1}{\sqrt{2}}(0, 0, 1, 1, 0).$$

19. Verify that u_1 and u_2 are orthonormal.
20. Let

$$x = (2, 4, 1, 3, 10).$$

Compute $E(x)$.

21. Compute $\hat{x} = D(E(x))$.
22. Which information about x is completely lost by the encoder?
23. Determine $\ker(E)$ explicitly in this example.
24. Check that the rank-nullity theorem holds.

Step 6 [Bonus] – Link with PCA

The previous construction becomes a simplified version of PCA when the directions

$$u_1, \dots, u_k$$

are chosen from the data so that they capture the most important directions of variation.

No prior knowledge of PCA is required here. For this exercise, you only need the following interpretation:

PCA chooses a low-dimensional subspace S so that projecting data onto S keeps as much variation as possible.

25. In the previous construction, what is the low-dimensional subspace?
26. What is the compressed representation of x ?
27. What is the reconstructed approximation of x ?
28. Explain why this procedure is linear dimensionality reduction.
29. Suppose $d = 10\,000$ and $k = 100$. How many coordinates are kept? What fraction of the original number of coordinates is this?
30. Explain why a low-dimensional representation can be useful for image retrieval or face recognition.

Step 7 [Bonus] – Deep image embeddings

In modern AI systems, image embeddings are often produced by neural networks. Such embeddings are usually nonlinear. Nevertheless, the same vector-space ideas remain important.

31. In an image retrieval system, one stores embeddings

$$z^{(1)}, \dots, z^{(m)} \in \mathbb{R}^k$$

for a database of images. Given a query image x_q , one computes $z_q = E(x_q)$. Explain how one could use these embeddings to retrieve visually similar images.

32. What is the main mathematical difference between the linear encoder E studied here and a neural-network image encoder?
33. Why is it still useful to understand the linear case before studying nonlinear embeddings?